
CLBench-life: Can Language Models Learn from Real-Life Context?

Abstract

Today’s AI assistants such as OpenClaw are designed to handle context effectively, making context learning an increasingly important capability for models. As these systems move beyond professional settings into everyday life, the nature of the contexts they handle also shifts. Real-life contexts are often messy, fragmented, and deeply tied to personal and social experience, such as multi-party conversations, personal archives, and behavioral traces. Yet it remains unclear whether current frontier language models can reliably learn from such contexts and solve tasks grounded in them. To this end, we introduce CLBench-life, a fully human-curated benchmark comprising 500 context-task pairs and 5,348 verification rubrics, covering common real-life scenarios. Solving tasks in CLBench-life requires models to reason over complex, messy real-life contexts, calling for strong real-life context learning abilities that go far beyond those evaluated in existing benchmarks. We evaluate ten frontier language models and find that real-life context learning remains highly challenging: even the best-performing model achieves only 19.3% task solving rate, while the average performance across models is only 13%. Models still struggle with contexts such as group chat histories and messy personal trails containing rich real-life behavioral traces. CLBench-life provides a crucial testbed for advancing real-life context learning, and progress on it can enable more intelligent and reliable AI assistants in everyday life.

1 Introduction

Real-world tasks often require models to search, organize, and reason over large amounts of context, rather than rely solely on pre-trained knowledge [89; 34; 76; 38]. This has driven the development of AI assistants such as OpenClaw [58] that excel at handling context. For the models that power these systems, the focus of improvement is accordingly shifting from enhancing their ability to reason over pre-trained knowledge to improving their ability to reason over complex context [2; 77; 53; 48]. The latter is commonly referred to as context learning, a foundational capability for handling complex context effectively [21; 43; 61].

At the same time, as AI assistants become more widely used, users increasingly expect them to handle everyday tasks [9; 80; 65]. Real-life contexts are typically less professional or domain-specific, yet far messier and more fragmented. For example, users may ask models to reason over discussions in a public thread or a meeting, or to organize and analyze personal digital traces and everyday behavioral records. However, existing benchmarks primarily evaluate models’ context learning ability in more professional and domain-specific contexts, such as finance [73; 27; 67], science [54; 12; 69; 40; 64], and code [4; 28; 11; 30]. They fail to capture whether current frontier language models (LMs) can reliably learn from real-life context. As a result, there is a pressing need for a benchmark that better reflects models’ real-life context learning ability and helps shape their future development.

To this end, we introduce CLBench-life, a benchmark for comprehensively evaluating context learning ability in real-life scenarios, comprising 500 expert-curated context-task pairs and 5,348 verification rubrics. Each task requires models to learn from complex, messy real-life contexts and reason over

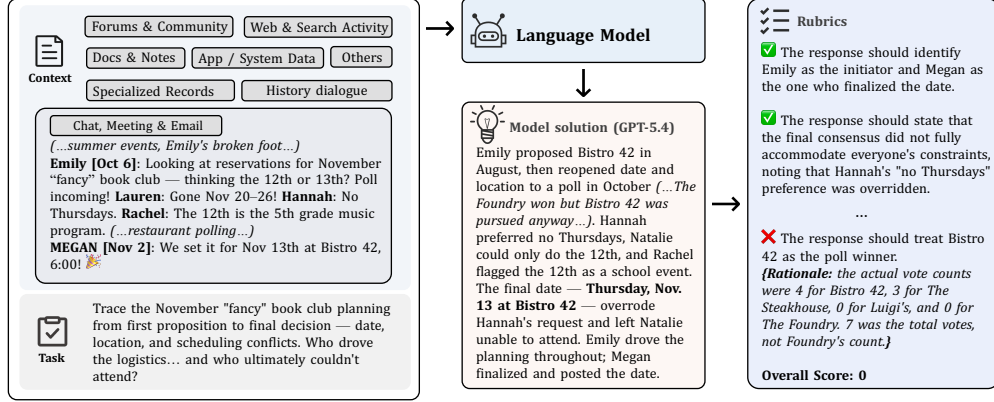


Figure 1: Each instance in CLBench-life consists of a task, a context, and a set of verification rubrics. The language model must reason over the provided context to solve the task, without retrieving additional external information. This example task in this figure requires reasoning over a long, informal group chat history to reconstruct an evolving real-life coordination process. The model must trace the planning of the November “fancy” book club meeting over time, resolve conflicting constraints, determine the final date and location, and identify who led and finalized the logistics.

new information, rather than merely relying on pre-trained knowledge. The context associated with each task is self-contained, so models do not need to retrieve additional external information to solve it. This allows CLBench-life to more directly evaluate context learning ability. CLBench-life contains three categories: communication & social interactions, fragmented information & revisions, and behavioral records & activity trails. The first category covers contexts such as private chats, group discussions, meeting transcripts, and public community interactions. The second focuses on fragmented information and revision processes, including personal notes, public information streams, and creation or revision histories. The third captures behavioral and activity-based records, such as game logs, digital footprints, browsing streams, and long-term records of daily activity. These categories cover common everyday scenarios and enable a broad evaluation of models’ real-life context learning ability.

We evaluate ten state-of-the-art language models on CLBench-life and find that real-life context learning remains highly challenging. On average, the models solve only 13% of tasks, and even the best-performing model, GPT-5.4, solves only 19.3%. Increasing reasoning effort generally improves context learning performance, but the extra token cost required to obtain similar gains varies substantially across models. We also find that task solve rate is not strongly correlated with context length, suggesting that poor real-life context learning cannot be simply attributed to limited long-context ability. Moreover, in group chat scenarios, models often exhibit confusion over user identities and references, leading them to misunderstand the context and fail to solve the task.

More detailed findings are presented in Sections 4 and 5. Overall, real-life context learning is a critical capability for models to handle everyday context effectively. Improving this ability can help enable more intelligent, practical, and capable AI assistants for real-life tasks. CLBench-life provides a rigorous benchmark for evaluating this capability and informing the future development of LMs in this direction.

2 Related Work

Real-world tasks are rarely solved in a single step and typically involve multi-step processes in which AI systems must continually acquire, organize, and utilize large amounts of complex contextual information [39; 89; 30; 78]. Accordingly, existing research can be broadly divided into two lines. One line focuses on the end-to-end capabilities of agents or AI assistants, such as how they leverage scaffolding to complete tasks [55; 88; 44]. The other line focuses on the context-related abilities of the underlying models themselves, such as whether they can effectively understand, organize, and learn from context [6; 8; 35; 25; 83; 1]. CLBench-life belongs more naturally to the latter line and further focuses on real-life context learning.

Benchmarks for agents. Many benchmarks aim to evaluate the overall capabilities of agents [41; 81; 72; 29; 16; 79; 13]. These benchmarks formulate tasks as multi-step processes of reasoning and acting, where models must continuously acquire, organize, and use contextual information through interactions with users and diverse tools, and then decide on the next action [51; 74; 80; 55]. In such settings, agent performance depends not only on task decomposition and tool use, but also on whether the underlying LM can effectively handle context and continue reasoning and acting based on that information [59]. However, these end-to-end benchmarks usually couple together many different abilities. When a model fails, it is often unclear whether the failure comes from tool use, memory retrieval, or the more fundamental ability of context learning itself [21; 10].

In contrast, CLBench-life provides a more direct and controlled evaluation of context learning as a standalone ability. It can be viewed as simulating the final step of agent problem solving: all task-relevant context is already provided, and the model is asked to solve the task based on that context alone. This largely removes errors introduced by earlier stages such as search or context organization, allowing context learning to be evaluated more directly.

Benchmarks for context-related abilities. Another line of work focuses more directly on the context-related abilities of LMs themselves [23; 43; 18; 86]. Some benchmarks evaluate long-context understanding, examining whether LMs can retrieve information and reason effectively under long-input settings [32; 1; 7; 19; 36; 82]. Unlike these benchmarks, CLBench-life is more concerned with whether models can learn from and reason over useful new information in messy real-life context and apply that information effectively. In fact, our results show that real-life context learning is not strongly correlated with conventional long-context ability, and the contexts in CLBench-life generally do not exceed the context windows of current frontier LMs.

Other benchmarks evaluate context learning more directly in professional domains, scientific discovery, and coding domains [66; 56; 21; 26]. CLBench-life is closely related to this line of work, but differs in the nature of its contexts. The contexts in CLBench-life are substantially messier, more fragmented, and much more grounded in everyday life.

There are also benchmarks that focus on context management or memory, evaluating whether models can store and retrieve historical information across long-term interactions [33; 70; 31; 57; 85]. CLBench-life differs from these settings in an important way: its tasks are solved in a one-shot setting, and all task-relevant new information has already been organized and provided to the model. Models do not need to call additional tools or perform retrieval. As a result, CLBench-life focuses less on how systems obtain context, and more on whether models can effectively learn from and use context once it is given.

Benchmarks for real-life scenarios. Beyond benchmarks built around professional or domain-specific tasks, evaluating models in real-life scenarios is also highly important [37]. Some benchmarks cover a wide range of settings, including long-term personal memory [42; 87; 68; 5; 75], multi-agent or collaborative scenarios [90; 49; 60; 50], multi-turn interactions with users and tools [24; 22; 51; 55], and everyday tasks such as web navigation or online shopping [79; 16; 89; 74].

Most of these benchmarks still evaluate agents in an end-to-end manner and do not explicitly disentangle the underlying abilities of LMs [84; 46]. In contrast, CLBench-life does not directly evaluate the overall performance of an agent in a full real-world environment. Instead, it evaluates context learning ability and, through this, indirectly reflects whether an agent can handle complex context when solving real-world tasks. In this sense, CLBench-life targets a more fundamental capability, enabling researchers to more clearly identify existing capability gaps and providing guidance for targeted model development.

3 CLBench-life

In this section, we first provide an overview of CLBench-life, then detail its context taxonomy and automatic evaluation method.

3.1 Overview

CLBench-life aims to evaluate whether language models can learn from real-life context and apply what they learn to solve grounded tasks. The evaluation pipeline is illustrated in Figure 1. Given

a context and a task, the model must generate a response based only on the provided information, and the solution is then automatically evaluated using task-level rubrics with a judge model. The context is self-contained, and all new information needed to solve a task is already organized into the provided context, so LMs do not need to use tools or retrieve external information.

Unlike existing context learning benchmarks that mainly focus on professional or domain-specific settings such as finance, science, and code [21; 26], CLBench-life centers on common everyday-life contexts. These contexts are diverse, ranging from multi-party conversations, personal notes, and archives to community discussions, behavioral logs, game histories, app or system data, web and search activity, and other forms of digital footprints. We present three cases with different types of context in Figure 2. Compared with professional-domain contexts, real-life contexts are often messier, more fragmented, and more personalized, which makes learning from it substantially more challenging.

The dataset statistics are reported in Table 1. Each instance in CLBench-life consists of a context, a task, and a set of task-level rubrics, all of which are manually curated. Moreover, about 59.8% of instances in CLBench-life are multi-turn. In these cases, the provided context also includes earlier rounds of interaction between the user and the assistant, while the final user turn specifies the target task to be solved. This increases task difficulty and more faithfully reflects how users interact with AI assistants in everyday scenarios.

3.2 Context Taxonomy

We categorize the contexts in CLBench-life into three broad types based on the kinds of information that people commonly deal with in everyday life. Each type is further divided into three subcategories, forming a taxonomy that captures the diversity of real-life contexts, as shown in Figure 3. Each category contains an equal number of contexts to avoid evaluation bias toward any single type. To illustrate the distinctive characteristics of the three types, we present one example case from each type in Figure 2.

Type 1: Communication & social interactions. This type covers contexts arising from interpersonal communication and social interaction in everyday life. Such contexts come from one-on-one conversations, multi-party group chats, meeting transcripts, or public community discussions such as forum threads, blog comments, and social media interactions. Compared with clean and well-organized text, they are often more informal, socially grounded, and messy. They often involve multiple participants, aliases or ambiguous references, topic shifts, interleaved threads, and nested replies. Relevant information is often scattered across long interactions rather than presented in a single coherent passage, making the context fragmented and difficult to follow.

This type is divided into three subcategories: (1) **private conversations**, which involve one-on-one interactions between individuals and often contain highly personal, informal, and topic-shifting exchanges; (2) **group conversations & meeting transcripts**, which contain multi-party discussions such as group chats, work meetings, or online voice-call transcripts; and (3) **community interactions**, which cover public-facing discussions in forums, comment sections, blog discussions, or social media threads.

To solve tasks in this category, models must reason task-relevant information from messy conversational context, track topic changes and interleaved discussion threads, resolve aliases and references, and reason about the relationships, intentions, and viewpoints of different participants.

Type 2: Fragmented information & revisions. This type covers contexts characterized by scattered, incomplete, or evolving pieces of information. These contexts are often drawn from notes, snippets,

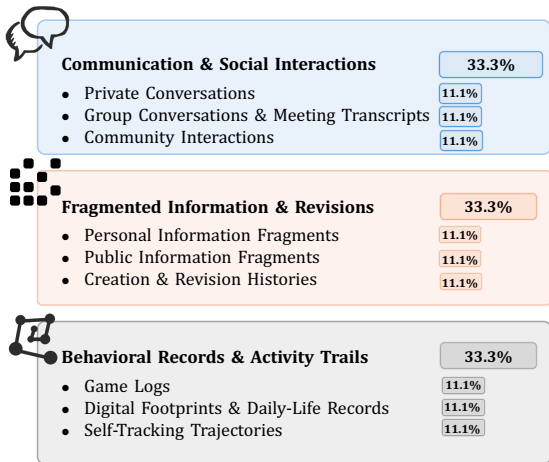


Figure 3: Context taxonomy of CLBench-life.



Figure 2: Three cases from CLBench-life. **Case 1:** A long, noisy group chat about planning a November "fancy" book club dinner. The task requires tracking changes in date and venue, reconciling conflicting member constraints, inferring final attendance from scattered updates, and judging whether the final plan satisfies the group's stated preferences. **Case 2:** Fragmented personal cycling archives, including trip plans, ride logs, weather disruptions, and maintenance notes. The task requires identifying trip-relevant evidence, inferring key failure modes, and producing a justified checklist for a future multi-day bike trip. **Case 3:** Workout logs collected before and after an injury break. The task requires comparing performance over time, inferring the most affected muscle groups, distinguishing recovery from plateau, and suggesting training adjustments.

Table 1: Statistics of CLBench-Life, including counts of tasks and rubrics, average user turns, rubrics per task, and context length in tokens.

Context Category	# Contexts	# Rubrics	Avg. User Turns	# Rubrics per Task			Context Length (tokens)		
				Min	Mean	Max	Min	Mean	Max
Communication & Social Interactions	135	1,812	1.9	2	13.4	36	5.6K	12.9K	34.9K
Fragmented Information & Revisions	135	1,877	2.0	3	13.9	38	5.4K	12.8K	34.4K
Behavioral Records & Activity Trails	135	1,659	1.8	2	12.3	40	6.8K	32.5K	170.8K
Total	405	5,348	1.9	2	13.2	40	5.4K	19.4K	170.8K

lists, archives, reading histories, or revision records. Unlike structured documents with a clear narrative or logical sequence, they are often loosely organized and fragmented. Relevant information may be spread across multiple partial records, mixed with outdated or less relevant content, or embedded in iterative edits and revisions.

This type is divided into three subcategories: **(1) personal information fragments**, which include personal notes, diaries, to-do lists, bookmarks, saved messages, and other private archives; **(2) public information fragments**, which contain scattered public information such as RSS feeds, news snippets, blog reading histories, search results, or public records; and **(3) creation & revision histories**, which capture iterative content-production processes such as document edit histories, collaborative comments, or revision suggestions.

Tasks in this category require models to organize and reason over loosely structured information, integrate evidence from multiple fragments, and resolve inconsistencies or version changes across records. For revision-heavy contexts in particular, models must also trace how content evolves and understand the intent behind successive changes. Overall, this category evaluates whether models can learn effectively from fragmented, evolving, and weakly structured context.

Type 3: Behavioral Records & Activity Trails. This type covers contexts composed of behavioral and activity-based records that document actions, movements, and habits over time. Rather than centering on dialogue or fragmented knowledge pieces, these contexts consist of raw activity traces, often recorded as logs, histories, timelines, or event sequences. They tend to be lengthy and highly structured, with task-relevant information often emerging from patterns across many records rather than from any single entry.

This type is divided into three subcategories: **(1) game logs**, which contain behavioral records from games or rule-based system, such as MOBA match replays, poker hand histories, and strategy execution logs; **(2) digital footprints & daily-life records**, which include browsing histories, transaction records, location logs, and other digital traces of everyday activity; and **(3) self-tracking trajectories**, which capture long-term self-monitoring data such as fitness logs, health records, learning progress, and other personal tracking histories.

Contexts in this category are also very common in real life. Tasks require models to reason over these real-life contexts, aggregate statistics, identify patterns, and provide useful insights. Depending on the subcategory, models may need to analyze strategy and game mechanics from logged events, infer routines and preferences from timestamped daily-life traces, or reason about trends, anomalies, and long-term changes in self-monitoring records. Overall, this category evaluates whether models can learn effectively from lengthy, behavior-centered, and record-intensive context.

3.3 Construction and automatic evaluation

Simplified construction process. CLBench-life is fully constructed by human annotators to ensure high data quality and to make the benchmark sufficiently challenging for frontier LMs. The simplified construction process for each instance involves the following steps¹. **(1)** We first defined and described common types of context that people encounter and deal with in everyday life through extensive discussion and investigation. **(2)** Given a specific context type, annotators then constructed a corresponding context by drawing from private sources, public sources, or newly created materials.

¹We provide a simplified description of the construction process due to policy restrictions.

Table 2: Task solving rate of ten frontier LMs on CLBench-Life. All models are evaluated in reasoning mode, with results reported as mean $\pm$ std (%) across three runs.

Model Names	Overall	Communication & Social interactions	Fragmented Information & Revisions	Behavioral Records & Activity Trails
GPT 5.4 (High)	19.3 $\pm$ 0.5	22.2 $\pm$ 0.7	15.8 $\pm$ 0.9	20.0 $\pm$ 0.7
Claude Opus 4.6 (High)	17.0 $\pm$ 1.3	21.7 $\pm$ 3.3	16.5 $\pm$ 0.9	12.8 $\pm$ 1.5
Gemini 3.1 Pro (High)	16.9 $\pm$ 1.2	18.8 $\pm$ 1.7	13.6 $\pm$ 1.9	18.3 $\pm$ 1.5
Seed 2.0 Pro (High)	15.5 $\pm$ 1.7	19.8 $\pm$ 2.3	12.1 $\pm$ 2.4	14.6 $\pm$ 1.1
Kimi K2.5 (High)	13.2 $\pm$ 0.8	15.3 $\pm$ 1.1	11.4 $\pm$ 1.5	12.8 $\pm$ 0.4
Qwen 3.5 Plus (High)	12.4 $\pm$ 0.1	13.1 $\pm$ 0.9	10.1 $\pm$ 1.1	14.1 $\pm$ 0.7
Grok 4.20 (High)	11.9 $\pm$ 0.4	12.8 $\pm$ 0.9	11.9 $\pm$ 1.5	10.9 $\pm$ 2.6
DeepSeek V3.2 Thinking	9.5 $\pm$ 0.4	12.3 $\pm$ 1.1	7.7 $\pm$ 1.1	8.6 $\pm$ 1.7
Hunyuan 2.0 Thinking	8.4 $\pm$ 0.7	9.4 $\pm$ 1.5	9.4 $\pm$ 1.5	6.4 $\pm$ 0.4
MiniMax M2.5	6.3 $\pm$ 1.0	8.6 $\pm$ 2.1	5.2 $\pm$ 1.3	5.2 $\pm$ 1.3

As described in Section 3.1, these contexts come from highly diverse sources. Sensitive information is removed from all contexts to avoid infringing on the rights and interests of others. (3) Annotators then designed sufficiently challenging tasks grounded in the context. These tasks are not intended to be simple needle-in-a-haystack retrieval problems or standard reading comprehension questions, nor can they be solved by relying only on pre-trained static knowledge. Instead, they evaluate models’ real-life context learning ability by requiring them to reason over the provided context and use newly learned information. (4) Finally, annotators write task-level rubrics for evaluation. These rubrics are required to be accurate, objective, and self-contained. They avoid introducing requirements that are not grounded in the task itself and avoid relying on subjective judgment whenever possible.

Moreover, supervisors also conduct multiple rounds of sampled quality inspection and feedback to ensure annotation quality in the annotation process. On average, annotating each context and its corresponding tasks requires approximately xx hours of expert effort.

Automatic evaluation. Complex real-life tasks in CLBench-life involve solutions that are difficult to check with predefined rules or may admit multiple valid solutions. To enable reliable automatic evaluation, we write task-level rubrics following prior work [21; 20]. Specifically, each rubric is formulated as a binary question that only allows a “yes” or “no” answer, where “yes” indicates that the model’s solution satisfies the rubric. On average, each task is associated with 10.7 rubrics.

In all experiments, we use GPT-5.1 with high reasoning effort as the verifier to evaluate model-generated solutions based on the rubrics. We find that introducing the original context and task during judging tends to increase instruction-following failures of the verifier and may also introduce subjective bias into the assessment of model solutions. Therefore, the rubrics in CLBench-life are decoupled from their corresponding contexts and tasks. For example, a rubric may take the form of “The response should correctly list the total gold for all ten participants at the five-minute mark: Participant 1 (1,350), P2 (1,753), ...” And during evaluation, we provide only the generated solution and the rubrics, without the original context or task. The system prompt used for the verifier follows Dou et al. [21].

We adopt a strict evaluation criterion: a model is considered to have successfully solved a task only if its solution passes all associated rubrics. We also analyze in Section 5.2 whether model rankings change under judges with different levels of strictness. To validate the reliability of this evaluation process, we randomly sample 100 model-generated solutions and their verification results and manually check them. The results show that the evaluation accuracy exceeds 90%. This observation is consistent with prior rubric-based evaluation work [17; 20], suggesting that the framework is reliable and rigorous for our benchmark. Moreover, in Table 6, we further investigate agreement across different judge models.

4 Benchmarking LMs with CLBench-life

Setup. We evaluate ten frontier language models on CLBench-life through their official APIs: GPT-5.4 from OpenAI [48], Claude-Opus-4.6 from Anthropic [3], Gemini-3.1-Pro from Google [14], Seed-2.0-Pro from ByteDance [62], Kimi-K2.5-Thinking from Moonshot [63], Qwen-3.5-Plus from

Table 3: Task solving rates across nine subcategories for all models. Columns are grouped by category. Cell shading indicates relative accuracy within each subcategory column, with darker cells indicating stronger performance.

Model	Communication & Social Interactions			Fragmented Information & Revisions			Behavioral Records & Activity Trails		
	Comm. Inter.	Group Conv.	Private Conv.	Pers. Info. Frag.	Pub. Info. Frag.	Create & Rev.	Game Logs	Digital FP & Daily	Self-Trk. Traj.
GPT-5.4 (High)	20.7	30.4	15.6	14.8	20.0	12.6	30.4	19.3	10.4
Claude Opus 4.6 (High)	16.3	28.1	20.7	14.8	14.8	20.0	17.8	13.3	7.4
Gemini 3.1 Pro (High)	10.4	28.9	17.0	10.4	11.9	18.5	34.1	15.6	5.2
Seed 2.0 Pro (High)	23.0	26.7	9.6	10.4	14.1	11.9	24.4	11.1	8.1
Kimi K2.5 (High)	9.6	23.7	12.6	11.1	9.6	13.3	17.8	11.9	8.9
Qwen 3.5 Plus (High)	13.3	14.8	11.1	11.1	7.4	11.9	20.7	15.6	5.9
Grok 4.20 (High)	10.4	16.3	11.9	13.3	12.6	9.6	15.6	13.3	3.7
DeepSeek V3.2 Thinking	7.4	17.0	12.6	8.1	8.1	6.7	12.6	7.4	5.9
Hunyuan 2.0 Thinking	5.9	9.6	12.6	9.6	11.1	7.4	6.7	6.7	5.9
MiniMax M2.5	4.4	10.4	11.1	6.7	4.4	4.4	5.2	7.4	3.0

Alibaba [52], Grok-4.20 from xAI [71], DeepSeek-V3.2-Thinking from DeepSeek [15], Hunyuan-2.0-Thinking from Tencent, and MiniMax-M2.5 [45]. For models that support explicit reasoning, we evaluate them under high reasoning effort. For each model, we run three independent trials and report the mean $\pm$ standard deviation. A task is counted as correct only when the model satisfies every rubric item. We use GPT-5.1 [47] in the high reasoning effort as the default judge to ensure grading reliability.

4.1 Main Results

Real-life context learning remains difficult for frontier LMs. As shown in Table 2, even the strongest model, GPT-5.4, achieves only a 19.3% task solving rate. Most models cluster between 8% and 17%, while Claude-Opus-4.6 and Gemini-3.1-Pro follow at 17.0% and 16.9%, respectively. No model surpasses 20%, indicating that learning from noisy, fragmented, and personal real-life context remains far from solved for current frontier models.

Task difficulty varies across context categories. Tasks in communication & social interactions are comparatively the easiest: GPT-5.4 reaches 22.2% and Claude-Opus-4.6 reaches 21.7%. By contrast, fragmented information & revisions is the hardest category, where the best score is only 16.5%. Models also exhibit distinct category-specific strengths. GPT-5.4 leads on behavioral records & activity trails as well as communication & social interactions, whereas Claude-Opus-4.6 performs best on fragmented information & revisions. Gemini-3.1-Pro is relatively strong on behavioral records & activity trails but underperforms on fragmented information & revisions tasks. Overall, frontier models show different strengths when processing different types of real-life context, rather than following a single consistent ranking across categories.

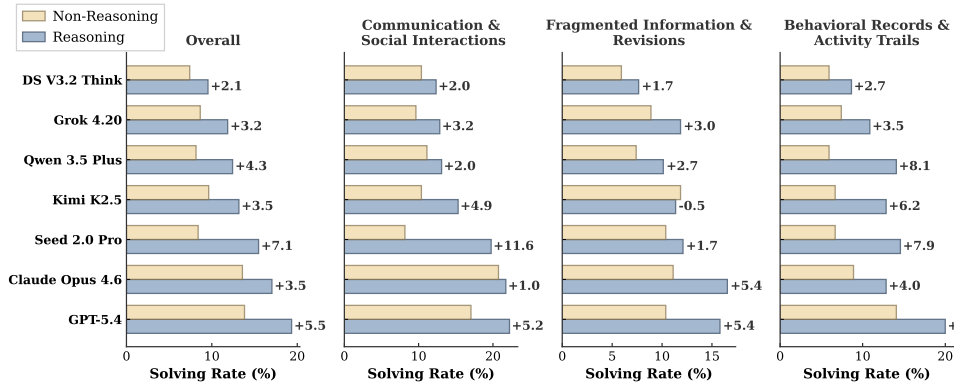


Figure 4: Performance comparison between reasoning and non-reasoning settings across seven models. Numbers next to the bars show the task solving rate gain obtained from reasoning. Reasoning improves overall performance for every model, with the largest and most consistent gains appearing on behavioral records & activity trails.

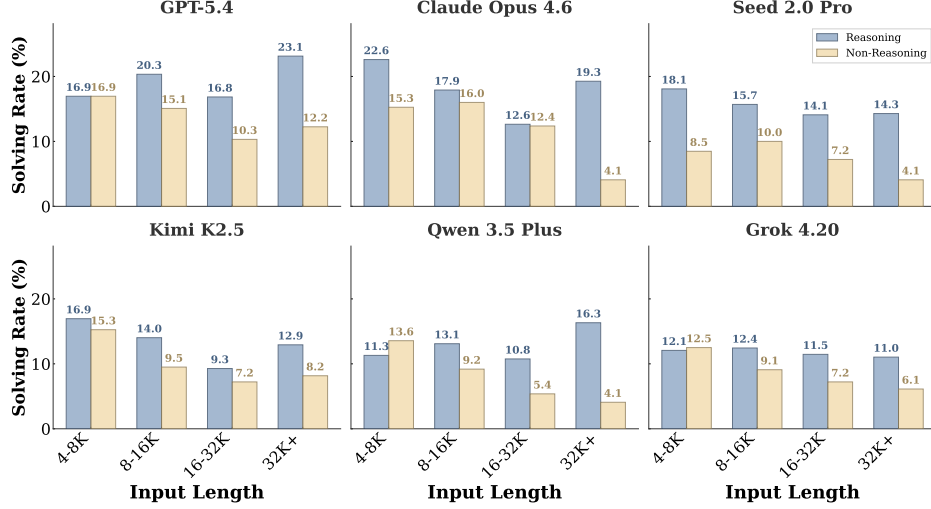


Figure 5: Task solving rate across context length bins under reasoning and non-reasoning settings.

Frontier models exhibit distinct strengths across fine-grained context categories. Table 3 breaks down performance across all nine subcategories. Group conversations & meeting transcripts and game logs are relatively tractable, with several top models reaching roughly 30% or higher. By contrast, self-tracking trajectories is consistently the hardest subcategory: the best score is only 10.4%, and most other models remain below 6%. Within fragmented information & revisions, GPT-5.4 is strongest on public information fragments, while Claude-Opus-4.6 and Gemini-3.1-Pro peak on creation & revision histories. Real-life context learning requires a diverse set of skills, including long-horizon behavioral synthesis, reconstruction from fragmented evidence, and socially grounded inference, all of which remain uneven in current models.

Real-life context learning benefits from explicit reasoning. Figure 4 compares reasoning and non-reasoning settings for seven models that support both. Enabling reasoning improves task solving rate for every model, with the largest and most consistent gains on behavioral records & activity trails. On this category, Qwen-3.5-Plus gains 8.1%, Seed-2.0-Pro gains 7.9%, and GPT-5.4 gains 5.9%, suggesting that reasoning is especially helpful for reconstructing behavioral patterns from scattered evidence. By contrast, gains on fragmented information & revisions and communication & social interactions are more uneven. On fragmented information & revisions, Claude-Opus-4.6 and GPT-5.4 both gain 5.4%, whereas Kimi-K2.5-Thinking shows a slight drop of 0.5%. On communication & social interactions, gains range from 1.0% for Claude-Opus-4.6 to 11.6% for Seed-2.0-Pro. Overall, CLBench-life tasks are designed to require strong, context-grounded inference over realistic everyday

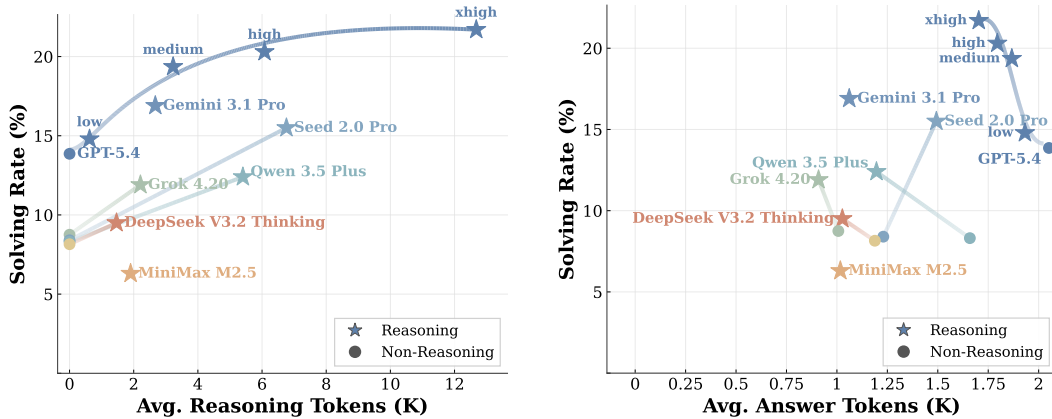


Figure 6: Average reasoning tokens (left) and answer tokens (right) versus task solving rate. Arrows connect reasoning and non-reasoning variants of the same model. More reasoning tokens generally correlate with higher solving rate. Longer answers do not reliably predict higher solving rate.

Table 4: Distribution of error types across models. The majority of solving failures are attributed to misusing knowledge in the context.

Model Names	Context Ignored (%)	Context Misused (%)	Format Error (%)	Refusal (%)
GPT-5.4 (High)	36.0	79.7	13.2	1.5
Claude Opus 4.6 (High)	38.9	76.0	16.2	0.9
Gemini 3.1 Pro (High)	44.4	77.8	14.3	2.3
Seed 2.0 Pro (High)	41.4	80.2	13.8	1.4
Kimi K2.5 (High)	45.3	80.2	12.7	1.1
Qwen 3.5 Plus (High)	45.1	76.9	13.0	2.0
Grok 4.20 (High)	40.0	83.7	12.4	2.5
Hunyuan 2.0 Thinking	42.8	80.1	10.6	0.8
DeepSeek V3.2 Thinking	39.1	83.1	15.3	1.6
MiniMax M2.5	43.5	84.4	13.4	0.8

material. Consistent with that design, enabling deeper reasoning improves performance, while the size of the gain still depends on both the model and the context category.

Real-life context learning is not reducible to long-context ability. The total input to language models consists of a context and a task specification, with the context accounting for most of the input length. Figure 5 shows task solving rate across context length bins under reasoning and non-reasoning settings. Without reasoning, several models exhibit clear degradation as context length increases, especially on the longest inputs. However, this pattern does not hold consistently with reasoning enabled. GPT-5.4 achieves its highest task solving rate of 23.3% on contexts above 32K, and Qwen-3.5-Plus peaks at 16.3% in the longest bin. Claude-Opus-4.6 and Kimi-K2.5-Thinking also recover on the longest inputs after declining in the middle ranges. These results indicate that input length alone does not determine task difficulty. The key challenge lies in whether models can effectively organize and reason over noisy, fragmented, and weakly structured real-life context.

Efficient reasoning is more valuable in real-life scenarios. Figure 6 breaks down output tokens into thinking tokens (left) and answer tokens (right) against task solving rate. The arrows connecting reasoning and non-reasoning variants of the same model reveal differences in *token efficiency*: how much performance improves per additional reasoning token. Seed-2.0-Pro turns 6.8K thinking tokens into a gain of 7.1 percentage points, while Qwen-3.5-Plus uses 5.4K tokens for a smaller gain of 4.3 percentage points. Gemini-3.1-Pro is particularly efficient, reaching 16.9% with only 2.7K thinking tokens, whereas MiniMax-M2.5 generates 1.9K thinking tokens yet still achieves the lowest task solving rate. These results show that simply producing more reasoning tokens does not guarantee better performance. On the answer-token side, shorter answers can still achieve higher task solving rate than longer ones. Our benchmark does not reward verbose or redundant outputs but favors responses that are accurate and directly targeted to the task. We argue that for practical AI assistants operating in real-life settings, reasoning efficiency matters more than sheer reasoning volume.

5 Further Analysis

5.1 Error Analysis

Context misuse is the primary failure mode. Table 4 presents the distribution of error types across models. Although both context ignored and context misused contribute substantially to model failures, context misused is consistently the dominant category. Its rate remains above 76% for every model, indicating that models often fail because they misunderstand or misuse the information in context. Meanwhile, context ignored rates vary from 36.0% to 45.3% and show a relationship with overall performance, with stronger models tending to ignore context less often. This pattern suggests that model capability improves contextual attention, but does not fully resolve the harder problem of correct contextual interpretation. Format errors are notably less common, at around 10% to 16%, and refusals remain rare at below 3%.

Role confusion is the dominant social context error in group conversations & meeting transcripts. To better understand model failures in socially grounded contexts, we conduct a focused error analysis

on group conversations & meeting transcripts tasks, where models must track multiple participants, their identities, their roles, and evolving information over time.

We identify four error sources. *Speaker Attribution*: attributing statements to the wrong person, *Alias Coreference*: failing to recognize that different names refer to the same person, *Role Confusion*: misidentifying who holds authority or responsibility, and *Stale Information*: relying on outdated information that was later corrected. As shown in Figure 7, Role Confusion is the dominant error category, with especially high counts for Kimi-K2.5-Thinking and Seed-2.0-Pro. Speaker Attribution is also frequent, appearing at comparable levels in Claude-Opus-4.6 and Gemini-3.1-Pro. Alias Coreference and Stale Information errors are less common. These results suggest that the difficulty of social context understanding largely arises from models struggling to maintain role structure and speaker-level distinctions throughout the conversation.

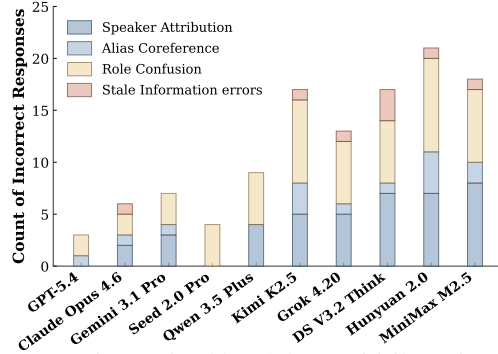


Figure 7: Fine-grained breakdown of failures in group conversations & meeting transcripts. Role confusion is the most prevalent error category overall, while speaker attribution is another common source of failure.

Table 5: Context ablation study. We strip all contexts from each task, retaining only the final user question, and evaluate GPT-5.4 under the same conditions. Performance drops from 19.3% to 1.7%, confirming that CLBench-Life tasks cannot be solved without the provided context.

Setting	Overall	Communication & Social interactions	Fragmented Information & Revisions	Behavioral Records & Activity Trails
With Context	19.3%	22.2%	15.8%	20.0%
Without Context	1.7%	2.2%	0.7%	2.2%
Δ	-17.6	-20.0	-15.1	-17.8

5.2 Evaluation Effectiveness Analysis

Solving CLBench-life tasks requires models to learn from the provided context. To verify that the benchmark cannot be solved from the final query alone, we conduct a no-context ablation in which all contexts are removed and only the user’s final task is retained. As reported in Table 5, GPT-5.4 drops from 19.3% to 1.7% when context is stripped away. The degradation is consistent across all categories. These near-zero results show that CLBench-life cannot be solved through parametric world knowledge alone, and instead requires models to acquire and use information from the provided real-life context.

Table 6: Inter-judge agreement on CLBench-life. We re-grade model outputs using two alternative judges, Claude-Opus-4.6 and Gemini-3.1-Pro, and compare their decisions against the default judge, GPT-5.1. All judge pairs achieve over 93% agreement, with Cohen’s κ values between 0.710 and 0.773, indicating substantial agreement beyond chance. Agreement rates and Cohen’s κ are averaged across five evaluated models.

Judge Pair	Agreement (%)	Cohen’s κ
GPT-5.1 vs Claude-Opus-4.6	93.2	0.724
GPT-5.1 vs Gemini-3.1-Pro	93.0	0.710
Claude-Opus-4.6 vs Gemini-3.1-Pro	94.5	0.773

Lenient pass-rate thresholds help distinguish partially solved cases from fully correct ones, while model rankings remain broadly stable across thresholds. We further analyze performance under varying rubric pass-rate thresholds rather than relying solely on binary pass/fail evaluation. As

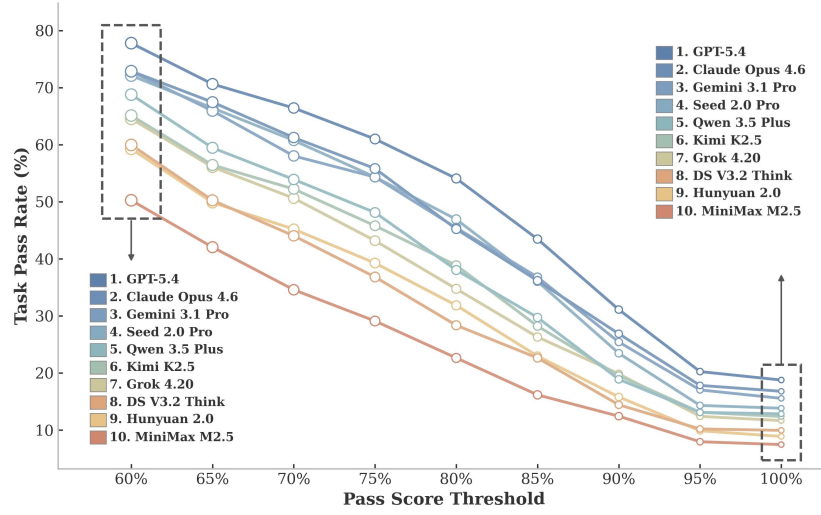


Figure 8: Task pass rate under different thresholds for all ten models. As the threshold increases from 60% to 100%, pass rates decrease monotonically for all models, showing that fully satisfying all rubrics is substantially harder than meeting only part of them. Stronger models remain consistently ahead across thresholds.

shown in Figure 8, pass rate decreases for all models as the threshold increases from 60% to 100%, indicating that many responses meet only part of the evaluation criteria under lenient settings, while fully satisfying all rubrics remains substantially more difficult. GPT-5.4 declines from about 77% at the 60% threshold to 19% at the 100% threshold. Meanwhile, model ranking remains broadly stable across thresholds, with stronger models maintaining an advantage under all criteria. In CLBench-life, each rubric is designed as a requirement that must be satisfied for a fully correct solution. This design makes the benchmark more sensitive to differences in answer quality, allowing it to distinguish partially correct responses from truly complete ones, while also yielding relatively stable evaluations across different scoring thresholds.

Rubric-based evaluation is robust across different judge models. We further assess the consistency of rubric-based evaluation by re-grading model outputs with different series models, Claude-Opus-4.6 and Gemini-3.1-Pro, and comparing their decisions against the default judge, GPT-5.1 (High). As shown in Table 6, pairwise agreement rates exceed 93% for all judge pairs, while Cohen’s κ values range from 0.710 to 0.773, indicating substantial agreement. These results suggest that our rubric-based evaluation is stable across judge models.

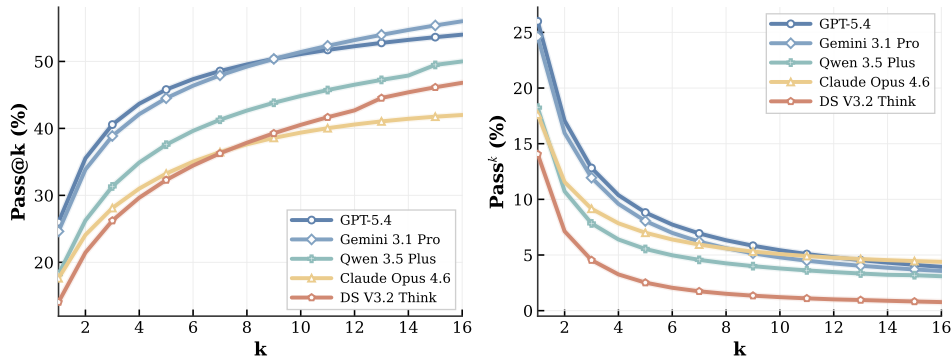


Figure 9: Pass@k and Pass^k curves on a subset of 50 CLBench-life questions. Pass@k measures the probability that at least one of k independent runs produces a correct answer; Pass^k measures the probability that all k runs are correct. Results are computed from up to 16 independent runs per model.

Pass^k reveals the difficulty of real-life context learning and the limited returns of inference-time scaling. To assess inference-time scaling, we evaluate five representative models with 16 independent runs each on a subset of 50 CLBench-life tasks. As shown in Figure 9, Pass@k increases with k for all models, but the gains saturate quickly, with most curves flattening after $k = 8$. This suggests that repeated sampling helps only on a limited set of borderline cases, while most failures reflect tasks that are intrinsically difficult for current models. Pass^k provides a complementary view of robustness. It declines sharply as k increases for every model, showing that correct answers are often not achieved consistently across runs. Even GPT-5.4 exhibits a substantial drop by $k = 4$, indicating that many tasks are solved only occasionally. Taken together, the rapid saturation of Pass@k and the steep decay of Pass^k suggest that the main bottleneck is not insufficient inference-time exploration, but the fundamental difficulty of real-life context learning itself.

Task uid: 17df0bd3-e99c-44ef-95fc-2400c8903cc9



System Prompt

Doc 1: March 2026 Bank Statement: Records all transactions for March. Includes purchases, transfers, and reversals

Doc 2: February 2026 Bank Statement: Lists all transactions for February. Highlights frequent 7-Eleven spending

Doc 3: January 2026 Bank Statement: Captures transactions across multiple merchants Spans several convenience store brands (7-Eleven, Circle K, Sinclair, Loaf n' Jug)

Doc 4: December 2025 Bank Statement: Provides earlier spending

Doc 5: Monthly Budget / Savings Plan Defines financial goals. Specifies March SoFi moving-savings goal: \$1,320



Task: I am trying to save money for moving expenses as I begin an apartment hunt for a 1 bedroom apt for just myself. Due to an injury on 2/13 I lost about 3 weeks of income. Look through my bank statements and monthly budgets and identify my biggest financial 'leak'; the one spending pattern (regular/repeated spending) that is currently holding me back from growing my savings? Cite 3 - 5 specific transactions to support your identified pattern. Include how much this 'pattern' is costing me financially by providing me a total spent from all 4 of the monthly statements, one total for each month, plus an Overall total that we can compare to my March SoFi Savings Goal."

Model solution (Gpt-4.5 (High))

Your biggest financial leak looks like: Convenience-store "quick stops" especially 7-Eleven/Circle K/Loaf N Jug.

Examples from your statements

12/16/2025 - ATM Withdrawal at 7ELEVEN-FCI: -\$100.00(more than 5)

Total cost of this pattern \n used a conservative total:- included convenience-store purchases \n- included ATM withdrawals clearly tied to those stops. December 2025 | \$136.22 | \n | January 2026 | \$333.05 | \n | February 2026 | \$209.84 | \n | March 2026 (through 3/2 only) | \$19.11 | \n | Overall Total | \$698.22

Compare that to your March SoFi moving-savings goal \n Your March SoFi savings goal is \$1,320.00. That is 52.9% of your March SoFi goal

Rubrics

- ✓ The response should identify that the primary financial "leak" consists of convenience spending, most often from convenience stores.
- ✗ The response should cite between 3 and 5 specific transactions from the provided documents that support the identified pattern.
- ✗ The response should report the monthly totals as: December 2025 — \$50.94; January 2026 — \$104.97; February 2026 — \$ 129.84; March 2026 — \$ 40.55.
- ✗ If the response includes transactions other than convenience store purchases, the response should calculate the overall total as the \$326.30 + totals from each of the other subcategories.
- ✓ The March SoFi Savings Goal is \$1,320.00

Figure 10: An example from CLBench-life where the task requires identifying the largest recurring financial leak from four months of bank statements. While GPT-5.4 (High) correctly identifies convenience store spending as the main leak, it fails to compute accurate monthly totals, misclassifies ATM withdrawals, and introduces unsupported speculative claims.

5.3 Qualitative Analysis

We select ten cases across multiple real-life context categories to gain deeper insights into model performance on context learning. These cases are drawn from GPT-5.4 (High), Claude-Opus-4.6 (High), Gemini-3.1-Pro (High), Seed-2.0-Pro, Kimi-K2.5-Thinking, and Qwen-3.5-Plus. We provide an in-depth analysis of all remaining examples in Appendix ??

In figure 10, we present an error case to illustrate the limitations of the current models. In this example, a user is planning to move, but has experienced income loss over the past three months due to an injury. She asks GPT-5.4 to identify the largest financial leak over the past four months, which means the recurring spending pattern that most significantly hinders her savings growth. She also provides more specific instructions: the model should (1) cite 3–5 concrete transactions as supporting evidence, (2) compute the monthly totals of this spending pattern across the four months as well as the overall total, and (3) compare this total against her March SoFi savings goal.

We then provide the user’s complete bank statements covering all income and expenses over the past four months, and prompt GPT-5.4 (high thinking) to generate a response. As shown in Figure X, the model successfully avoids being distracted by other spending categories (e.g., gambling-related transactions, subscription charges, etc.), and correctly identifies convenience store spending as the primary financial leak. It also demonstrates strong contextual understanding by locating the user’s March SoFi savings goal and incorporating it into a comparative analysis.

However, the model makes critical numerical and reasoning errors. Specifically, the monthly totals for convenience store spending are all incorrect. For example, the true total for December is 50.94,

but the model reports 136.22. A key source of error is that the model incorrectly classifies ATM withdrawals at convenience stores as part of convenience store spending. Even though the rubric explicitly accounts for this possibility and provides clear instructions on how such cases should be handled, the model still fails to compute the totals correctly. Furthermore, to mitigate hallucination, the user explicitly requires that all conclusions must be grounded in the provided documents, without speculative reasoning. Nevertheless, the model introduces unsupported claims such as “likely under-budgeted or invisible” and “a strong sign this habit is draining money,” which go beyond the evidence available in the bank statements.

Across all analyzed cases, we observe several consistent limitations of current models. First, instruction-following remains unreliable: even when explicit requirements on calculation, evidence citation, and grounding are clearly specified, models frequently produce fluent but non-compliant responses. Second, models struggle with precise aggregation and structured reasoning over long, messy real-life context, often making systematic numerical or logical errors despite correctly identifying high-level patterns. Third, grounding remains incomplete, with models occasionally introducing unsupported or speculative claims beyond the provided context. Overall, these findings suggest that current frontier models still lack robustness in real-life context learning, particularly in tasks requiring accurate computation, strict adherence to instructions, and fully evidence-grounded reasoning.

6 Discussion

Limitations and future work. CLBench-life has several limitations that point to important directions for future work. First, group chat emerges as a particularly important yet challenging real-life setting. Models often struggle with phenomena such as alias and reference confusion in long multi-user interaction histories, suggesting the need for more targeted research on understanding and reasoning over group chat context. Second, CLBench-life currently focuses on text-only context. However, real-life context learning is inherently multimodal, and extending the benchmark to incorporate visual context is an important next step. Thirdly, our rubric-based evaluation currently relies on GPT-5.1 as the judge model, which is effective but costly. Meanwhile, we observe that smaller or open-source judge models exhibit noticeable bias. Designing low-cost and reliable rubric-based evaluation frameworks remains an open challenge. Finally, we do not provide details of the construction process of CLBench-life due to policy restrictions.

Conclusion. We present CLBench-life, the first benchmark for evaluating context learning in everyday real-life scenarios. CLBench-life is fully human-annotated and covers common real-life settings. The contexts in it are sufficiently messy and complex to pose a meaningful challenge to current SOTA language models. Our evaluations show that real-life context learning remains a major challenge for frontier LMs, with substantial room for improvement. In addition to overall performance gaps, we uncover several important findings about the relationships between performance and reasoning effort, solution length, context length, and long-context ability. We hope CLBench-life will provide a strong foundation for future research on real-life context learning and help drive the development of more intelligent and practical AI assistants for everyday use.

Acknowledgments

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